1

CLASSMATE
Date :
Page :

MINI PROJECT USING JAVA

Assignment no - 03

Problem statement -

Write a Java program to impliment method overloading and method overriding in Java.

1. Write a class shape. Find the area of basic shape using polymorphism (constructor overloading and method overloading)
2. Write a Java program to create a parent class Hillstations and three subclasses by hill station names. Subclass extend the superclass and override methods famousfood() and famousfor() Call the method famousfood() and famousfor() by the parent class, i.e. Hillstations class. It should refer to the base class object and the base class method overrides the superclass method the superclass method, base class method is invoked at runtime.

Objective -

1. To study method overloading
2. To study method overloading in Java

Theory -

1. Method Overloading and its advantages.

is a features in Java where multiple methods have the same name but different parameters (type, number or order)

Eg - class shapes {
```
        int area(int side) {
                return side * side; }
      int area (int length, int breadth) {
            return length * breadth; }
```

Advantages -
- Improves code readability
- Supports compile-time polymorphism
- Eliminates the need for different method names
- Makes code more flexible and reusable

2. Method Overriding and its Rule.
occurs when a subclass provides a specific implementation of a method that is already defined in its parent class

eg -
```
class Hillstation {
    void famousfood() {
        System.out.println("General Hill food");
}}
class Manali extends HillStations {
    void famousford() {
        System.out.println("Siddu");
}}
```

Rules of Method Overriding
- Method must have the same name, parameters & return type
- Must be in parent-child relationship (inheritance)
- Cannot override final, static or private method
- Access modifier should be same or less restrictive
- Supports runtime polymorphism.

Conclusion -
Thus, we have successfully implemented method overloading and method overriding in Java

FAQs -

Q1 Can we overload java main() method?
Yes, we can overload the main() method by changing its parameters. However, JVM will only call:
public static void main(String [ ] args)

Q2 Can we declare overloaded methods as final?
Yes, overload methods can be declare as final
But once marked final, they cannot be overridden in subclasses.

Q3 Can overloaded methods be overridden?
Yes, A method that is overloaded in a class can still be overridden in a subclass if it follows overriding rules

Q4. What is method overriding?
Method overriding is redifining a method in a subclass with the same signature as in the parent class to provide a specific implementation.

Q5 What happens if we change the arguments of an overriding method?
If arguments are changed, it is not overriding anymore - it becomes method overloading.

Q6 Can we change the return type of an overriding method from number type to integer type?
Yes, but only if it is a covariant return type.
For example:
- Parent method returns Number.
- Child method can return integer (since integer is a subclass)